3.3.16 Chromatography (A-level only)

Chromatography provides an important method of separating and identifying components in a mixture. Different types of chromatography are used depending on the composition of mixture to be separated.

Content	Opportunities for skills development
Chromatography can be used to separate and identify the components in a mixture. Types of chromatography include: • thin-layer chromatography (TLC) – a plate is coated with a solid and a solvent moves up the plate • column chromatography (CC) – a column is packed with a solid and a solvent moves down the column • gas chromatography (GC) – a column is packed with a solid or with a solid coated by a liquid, and a gas is passed through the column under pressure at high temperature. Separation depends on the balance between solubility in the moving phase and retention by the stationary phase. Retention times and R_f values are used to identify different substances. The use of mass spectrometry to analyse the components separated by GC. **Students should be able to:** • calculate R_f values from a chromatogram • compare retention times and R_f values with standards to identify different substances.	**AT a, i and k** **PS 1.2, 3.2 and 4.1** Students could use thin-layer chromatography to identify analgesics. Students could use thin-layer chromatography to identify transition metal ions in a solution.
Required practical 12 Separation of species by thin-layer chromatography.	